Orbital Debris EDA

This project analyzes orbital debris and satellite catalogs to support a collision-avoidance simulation environment.

Primary objectives:

- profile debris and satellite populations
- clean and standardize source datasets
- produce evidence-driven insights for environment design and evaluation

Datasets used:

- CelesTrak SATCAT
- UCS Satellite Database
- ESA Kelvins "Space Debris: the origin"

Data Loading and Initial Inspection

Load the raw datasets, verify file availability, and inspect basic structure (shape, columns, sample rows).

```
from pathlib import Path
import pandas as pd

# Base paths
ROOT = Path.cwd()
RAW_DIR = ROOT / "input_data" / "raw"

# Files used in this analysis
SATCAT_PATH = RAW_DIR / "satcat.csv"
UCS_PATH = RAW_DIR / "satellites.csv"
KELVINS_DIR = RAW_DIR / "space-debris-the-origin"
KELVINS_LABELS_PATH = KELVINS_DIR / "labels_train.dat"

print("Root:", ROOT)
print("Raw data dir:", RAW_DIR)

# 1) Existence checks
required = {
    "satcat": SATCAT_PATH,
    "ucs": UCS_PATH,
    "kelvins_dir": KELVINS_DIR,
    "kelvins_labels": KELVINS_LABELS_PATH,
}

print("\nFile checks:")
```

```
for name, path in required.items():
    print(f"- {name}: {'OK' if path.exists() else 'MISSING'} ->
{path}")

# 2) Read SATCAT
satcat_df = pd.read_csv(SATCAT_PATH)
print("\nSATCAT shape:", satcat_df.shape)
print("SATCAT columns:", list(satcat_df.columns))
print("SATCAT preview:")
display(satcat_df.head(3))

# 3) Read UCS
# Many UCS exports are semicolon-delimited; we load with ';' first and
standardize later.
ucs_df = pd.read_csv(UCS_PATH, sep=";", engine="python")
print("\nUCS shape:", ucs_df.shape)
print("UCS columns (first 15):", list(ucs_df.columns[:15]))
print("UCS preview:")
display(ucs_df.head(3))

# 4) Read Kelvins labels (whitespace-delimited, no header)
# Column meaning from challenge docs:
#   0: originator satellite id
#   1: area-to-mass ratio
#   2..8: epoch/orbital-element values
kelvins_labels = pd.read_csv(KELVINS_LABELS_PATH, sep=r"\s+",
header=None)
print("\nKelvins labels shape:", kelvins_labels.shape)
print("Kelvins labels preview:")
display(kelvins_labels.head(5))
```

```
Root: /home/mayank/repos/orbits
Raw data dir: /home/mayank/repos/orbits/input_data/raw

File checks:
- satcat: OK -> /home/mayank/repos/orbits/input_data/raw/satcat.csv
- ucs: OK -> /home/mayank/repos/orbits/input_data/raw/satellites.csv
- kelvins_dir: OK -> /home/mayank/repos/orbits/input_data/raw/space-
debris-the-origin
- kelvins_labels: OK ->
/home/mayank/repos/orbits/input_data/raw/space-debris-the-origin/label
s_train.dat

SATCAT shape: (68330, 17)
SATCAT columns: ['OBJECT_NAME', 'OBJECT_ID', 'NORAD_CAT_ID',
'OBJECT_TYPE', 'OPS_STATUS_CODE', 'OWNER', 'LAUNCH_DATE',
'LAUNCH_SITE', 'DECAY_DATE', 'PERIOD', 'INCLINATION', 'APOGEE',
'PERIGEE', 'RCS', 'DATA_STATUS_CODE', 'ORBIT_CENTER', 'ORBIT_TYPE']
SATCAT preview:
```

```
  OBJECT_NAME  OBJECT_ID  NORAD_CAT_ID OBJECT_TYPE OPS_STATUS_CODE
OWNER  \
0    SL-1 R/B  1957-001A             1         R/B               D
CIS
1   SPUTNIK 1  1957-001B             2         PAY               D
CIS
2   SPUTNIK 2  1957-002A             3         PAY               D
CIS

  LAUNCH_DATE LAUNCH_SITE  DECAY_DATE  PERIOD  INCLINATION  APOGEE
PERIGEE  \
0  1957-10-04       TYMSC  1957-12-01   96.19        65.10   938.0
214.0
1  1957-10-04       TYMSC  1958-01-03   96.10        65.00  1080.0
64.0
2  1957-11-03       TYMSC  1958-04-14  103.74        65.33  1659.0
211.0

     RCS DATA_STATUS_CODE ORBIT_CENTER ORBIT_TYPE
0  20.42              NaN           EA        IMP
1    NaN              NaN           EA        IMP
2   0.08              NaN           EA        IMP


UCS shape: (7563, 68)
UCS columns (first 15): ['Name of Satellite, Alternate Names',
'Current Official Name of Satellite', 'Country/Org of UN Registry',
'Country of Operator/Owner', 'Operator/Owner', 'Users', 'Purpose',
'Detailed Purpose', 'Class of Orbit', 'Type of Orbit', 'Longitude of
GEO (degrees)', 'Perigee (km)', 'Apogee (km)', 'Eccentricity',
'Inclination (degrees)']
UCS preview:

                  Name of Satellite, Alternate Names  \
0  1HOPSAT-TD (1st-generation High Optical Perfor...
1                            AAC AIS-Sat1 (Kelpie 1)
2                                            Aalto-1

  Current Official Name of Satellite Country/Org of UN Registry  \
0                          1HOPSAT-TD                        NR
1            AAC AIS-Sat1 (Kelpie 1)             United Kingdom
2                            Aalto-1                    Finland

  Country of Operator/Owner    Operator/Owner       Users  \
0                       USA      Hera Systems  Commercial
1            United Kingdom   AAC Clyde Space  Commercial
2                   Finland  Aalto University       Civil

                  Purpose                       Detailed Purpose  \
0       Earth Observation                       Infrared Imaging
```

```
1       Earth Observation  Automatic Identification System (AIS)
2  Technology Development                                     NaN

  Class of Orbit       Type of Orbit  ... Unnamed: 58  Unnamed: 59  \
0            LEO  Non-Polar Inclined  ...         NaN          NaN
1            LEO     Sun-Synchronous  ...         NaN          NaN
2            LEO     Sun-Synchronous  ...         NaN          NaN

  Unnamed: 60 Unnamed: 61 Unnamed: 62 Unnamed: 63 Unnamed: 64 Unnamed:
65  \
0         NaN         NaN         NaN         NaN         NaN
NaN
1         NaN         NaN         NaN         NaN         NaN
NaN
2         NaN         NaN         NaN         NaN         NaN
NaN

  Unnamed: 66 Unnamed: 67
0         NaN         NaN
1         NaN         NaN
2         NaN         NaN

[3 rows x 68 columns]


Kelvins labels shape: (100, 9)
Kelvins labels preview:

    0         1       2         3         4        5         6
7  \
0  94  10.32583 -3815.0  42260.01  0.047700  11.3705   25.6240
91.5576
1  17  20.08288 -3845.0  42260.66  0.008111   8.2732   56.3063
272.0286
2  18   0.52148 -7275.0  42252.96  0.099613   2.2224   13.8162
242.1460
3  73  41.77603 -4825.0  42258.87  0.057331  15.0011  237.5817
299.0621
4  10   3.60228 -6355.0  42249.90  0.091040   6.5579  206.5880
286.8121

          8
0    9.5178
1   45.9370
2  166.9770
3   72.5915
4  329.0498
```

UCS Data Preparation

Standardize UCS fields and save cleaned output to `input_data/processed/ucs_clean.csv`.

```
# Create a working copy of UCS data
ucs_clean = ucs_df.copy()

# Ensure processed output directory exists
PROCESSED_DIR = ROOT / "input_data" / "processed"
PROCESSED_DIR.mkdir(parents=True, exist_ok=True)

# 1) Normalize header spacing
ucs_clean.columns = [c.strip() for c in ucs_clean.columns]

# 2) Drop parser-generated placeholder columns from the raw export
# These often appear as Unnamed:* after parsing rows with trailing
separators.
unnamed_cols = [c for c in ucs_clean.columns if
c.startswith("Unnamed:")]
ucs_clean = ucs_clean.drop(columns=unnamed_cols, errors="ignore")

# 3) Split combined first field: "Name of Satellite, Alternate Names"
# into two explicit columns where possible.
first_col = "Name of Satellite, Alternate Names"
if first_col in ucs_clean.columns:
    split_cols = ucs_clean[first_col].astype(str).str.split(",", n=1,
expand=True)
    ucs_clean["Name of Satellite"] = split_cols[0].str.strip()
    ucs_clean["Alternate Names (raw)"] =
split_cols[1].fillna("").str.strip()

# 4) Standardize key numeric-like columns that may contain decimal
commas
numeric_candidates = [
    "Perigee (km)",
    "Apogee (km)",
    "Eccentricity",
    "Inclination (degrees)",
    "Period (minutes)",
    "Launch Mass (kg.)",
    "Dry Mass (kg.)",
    "Power (watts)",
]

for col in numeric_candidates:
    if col in ucs_clean.columns:
        ucs_clean[col] = (
            ucs_clean[col]
            .astype(str)
```

```
            .str.replace(",", ".", regex=False)
            .str.replace(" ", "", regex=False)
        )
        ucs_clean[col] = pd.to_numeric(ucs_clean[col],
errors="coerce")

# 5) Parse launch date
if "Date of Launch" in ucs_clean.columns:
    ucs_clean["Date of Launch"] = pd.to_datetime(
        ucs_clean["Date of Launch"], errors="coerce", dayfirst=True,
format="mixed"
    )

# 6) Persist cleaned dataset
ucs_clean_path = PROCESSED_DIR / "ucs_clean.csv"
ucs_clean.to_csv(ucs_clean_path, index=False)

print("Original UCS shape:", ucs_df.shape)
print("Cleaned UCS shape:", ucs_clean.shape)
print("Dropped unnamed columns:", len(unnamed_cols))
print("Saved cleaned file:", ucs_clean_path)

print("\nCleaned columns (first 25):")
print(list(ucs_clean.columns[:25]))

print("\nMissing values (top 10 columns):")
print(ucs_clean.isna().sum().sort_values(ascending=False).head(10))

display(ucs_clean.head(3))
```

```
Original UCS shape: (7563, 68)
Cleaned UCS shape: (7563, 38)
Dropped unnamed columns: 32
Saved cleaned file:
/home/mayank/repos/orbits/input_data/processed/ucs_clean.csv

Cleaned columns (first 25):
['Name of Satellite, Alternate Names', 'Current Official Name of
Satellite', 'Country/Org of UN Registry', 'Country of Operator/Owner',
'Operator/Owner', 'Users', 'Purpose', 'Detailed Purpose', 'Class of
Orbit', 'Type of Orbit', 'Longitude of GEO (degrees)', 'Perigee (km)',
'Apogee (km)', 'Eccentricity', 'Inclination (degrees)', 'Period
(minutes)', 'Launch Mass (kg.)', 'Dry Mass (kg.)', 'Power (watts)',
'Date of Launch', 'Expected Lifetime (yrs.)', 'Contractor', 'Country
of Contractor', 'Launch Site', 'Launch Vehicle']

Missing values (top 10 columns):
Power (watts)       7335
Source.6            7060
Source.5            7010
```

```
Source.1            6838
Source.4            6835
Dry Mass (kg.)      6799
Source.3            6438
Detailed Purpose    6308
Source.2            5732
Comments            5478
dtype: int64

                   Name of Satellite, Alternate Names  \
0  1HOPSAT-TD (1st-generation High Optical Perfor...
1                            AAC AIS-Sat1 (Kelpie 1)
2                                             Aalto-1

  Current Official Name of Satellite Country/Org of UN Registry  \
0                          1HOPSAT-TD                        NR
1            AAC AIS-Sat1 (Kelpie 1)             United Kingdom
2                            Aalto-1                    Finland

  Country of Operator/Owner    Operator/Owner       Users  \
0                       USA      Hera Systems  Commercial
1            United Kingdom   AAC Clyde Space  Commercial
2                   Finland  Aalto University       Civil

                   Purpose                        Detailed Purpose  \
0       Earth Observation                        Infrared Imaging
1       Earth Observation  Automatic Identification System (AIS)
2  Technology Development                                     NaN

  Class of Orbit       Type of Orbit  ... Source Used for Orbital Data
\
0            LEO  Non-Polar Inclined  ...                JMSatcat/3_20

1            LEO     Sun-Synchronous  ...                JMSatcat/9_23

2            LEO     Sun-Synchronous  ...               JMSatcat/10_17


                                                 Source  \
0  https://spaceflig htnow.com/2019/12/11/indias-...
1  https://www.aac-clyde.space/articles/aac-clyde...
2  https://directory.eoportal.org/web/eoportal/sa...

                    Source.1
Source.2  \
0  https://www.herasys.com/
NaN
1                       NaN
NaN
2                       NaN
```

```
http://www.planet4589.org/space/log/satcat.txt

   Source.3  Source.4  Source.5  Source.6  \
0       NaN       NaN       NaN       NaN
1       NaN       NaN       NaN       NaN
2       NaN       NaN       NaN       NaN

                                   Name of Satellite Alternate Names
(raw)
0  1HOPSAT-TD (1st-generation High Optical Perfor...

1                             AAC AIS-Sat1 (Kelpie 1)

2                                                Aalto-1


[3 rows x 38 columns]
```

SATCAT Preparation

Standardize SATCAT fields and save a cleaned table to `input_data/processed/satcat_clean.csv`.

```
satcat_clean = satcat_df.copy()

for date_col in ["LAUNCH_DATE", "DECAY_DATE"]:
    if date_col in satcat_clean.columns:
        satcat_clean[date_col] =
pd.to_datetime(satcat_clean[date_col], errors="coerce")

numeric_satcat = ["PERIOD", "INCLINATION", "APOGEE", "PERIGEE", "RCS"]
for col in numeric_satcat:
    if col in satcat_clean.columns:
        satcat_clean[col] = pd.to_numeric(satcat_clean[col],
errors="coerce")

satcat_clean_path = PROCESSED_DIR / "satcat_clean.csv"
satcat_clean.to_csv(satcat_clean_path, index=False)

print("SATCAT original:", satcat_df.shape)
print("SATCAT cleaned:", satcat_clean.shape)
print("Saved:", satcat_clean_path)
display(satcat_clean.head(3))
```

```
SATCAT original: (68330, 17)
SATCAT cleaned: (68330, 17)
Saved: /home/mayank/repos/orbits/input_data/processed/satcat_clean.csv

  OBJECT_NAME  OBJECT_ID  NORAD_CAT_ID OBJECT_TYPE OPS_STATUS_CODE
OWNER  \
```

```
0    SL-1 R/B  1957-001A             1         R/B                D
CIS
1   SPUTNIK 1  1957-001B             2         PAY                D
CIS
2   SPUTNIK 2  1957-002A             3         PAY                D
CIS

  LAUNCH_DATE LAUNCH_SITE DECAY_DATE  PERIOD  INCLINATION  APOGEE
PERIGEE  \
0  1957-10-04       TYMSC 1957-12-01   96.19        65.10   938.0
214.0
1  1957-10-04       TYMSC 1958-01-03   96.10        65.00  1080.0
64.0
2  1957-11-03       TYMSC 1958-04-14  103.74        65.33  1659.0
211.0

     RCS DATA_STATUS_CODE ORBIT_CENTER ORBIT_TYPE
0  20.42              NaN           EA        IMP
1    NaN              NaN           EA        IMP
2   0.08              NaN           EA        IMP
```

Kelvins Preparation

Parse labels and trajectory files, then save processed outputs under `input_data/processed`.

```
from pathlib import Path

kelvins_labels_clean = kelvins_labels.copy()
kelvins_labels_clean.columns = [
    "originator_id",
    "area_to_mass_ratio",
    "feature_2",
    "feature_3",
    "semi_major_axis_km",
    "eccentricity",
    "inclination_deg",
    "mean_anomaly_deg",
    "raan_or_arg_deg",
]

kelvins_labels_path = PROCESSED_DIR / "kelvins_labels_clean.csv"
kelvins_labels_clean.to_csv(kelvins_labels_path, index=False)


def read_kelvins_folder(folder_path: Path):
    files = sorted(folder_path.glob("*.dat"))
    frames = []
    for file in files:
        temp = pd.read_csv(file, sep=r"\s+", header=None)
```

```
        temp["trajectory_file"] = file.name
        temp["trajectory_id"] = file.stem
        frames.append(temp)
    if not frames:
        return pd.DataFrame()
    return pd.concat(frames, ignore_index=True)

kelvins_deb_train = read_kelvins_folder(KELVINS_DIR / "deb_train")
kelvins_deb_test = read_kelvins_folder(KELVINS_DIR / "deb_test")
kelvins_sat = read_kelvins_folder(KELVINS_DIR / "sat")

kelvins_deb_train.to_csv(PROCESSED_DIR / "kelvins_deb_train_long.csv",
index=False)
kelvins_deb_test.to_csv(PROCESSED_DIR / "kelvins_deb_test_long.csv",
index=False)
kelvins_sat.to_csv(PROCESSED_DIR / "kelvins_sat_long.csv",
index=False)

print("Kelvins labels shape:", kelvins_labels_clean.shape)
print("Kelvins deb_train shape:", kelvins_deb_train.shape)
print("Kelvins deb_test shape:", kelvins_deb_test.shape)
print("Kelvins sat shape:", kelvins_sat.shape)
print("Saved label file:", kelvins_labels_path)
```

```
Kelvins labels shape: (100, 9)
Kelvins deb_train shape: (751, 9)
Kelvins deb_test shape: (751, 9)
Kelvins sat shape: (109600, 9)
Saved label file:
/home/mayank/repos/orbits/input_data/processed/kelvins_labels_clean.cs
v
```

Data Quality Summary

Create compact quality tables for cleaned SATCAT, cleaned UCS, and Kelvins labels.

```
def quality_table(df: pd.DataFrame, name: str) -> pd.DataFrame:
    return pd.DataFrame(
        {
            "dataset": [name],
            "rows": [len(df)],
            "columns": [df.shape[1]],
            "duplicate_rows": [int(df.duplicated().sum())],
            "missing_cells": [int(df.isna().sum().sum())],
            "missing_pct": [round(df.isna().sum().sum() / (df.shape[0]
* df.shape[1]) * 100, 2)],
        }
    )
```

```
quality_overview = pd.concat(
    [
        quality_table(satcat_clean, "satcat_clean"),
        quality_table(ucs_clean, "ucs_clean"),
        quality_table(kelvins_labels_clean, "kelvins_labels_clean"),
    ],
    ignore_index=True,
)

display(quality_overview)

missing_by_dataset = {
    "satcat_clean":
satcat_clean.isna().sum().sort_values(ascending=False).head(10),
    "ucs_clean":
ucs_clean.isna().sum().sort_values(ascending=False).head(10),
    "kelvins_labels_clean":
kelvins_labels_clean.isna().sum().sort_values(ascending=False).head(10
),
}

for name, series in missing_by_dataset.items():
    print(f"\nTop missing columns in {name}:")
    print(series)
```

```
                dataset   rows  columns  duplicate_rows  missing_cells
\
0          satcat_clean  68330       17               0         156098

1             ucs_clean   7563       38               7          74207

2  kelvins_labels_clean    100        9               0              0


   missing_pct
0        13.44
1        25.82
2         0.00


Top missing columns in satcat_clean:
DATA_STATUS_CODE    67072
RCS                 35399
DECAY_DATE          33383
OPS_STATUS_CODE     16560
INCLINATION           921
APOGEE                921
PERIOD                921
PERIGEE               921
OBJECT_NAME             0
```

```
LAUNCH_SITE              0
dtype: int64

Top missing columns in ucs_clean:
Power (watts)       7335
Source.6            7060
Source.5            7010
Source.1            6838
Source.4            6835
Dry Mass (kg.)      6799
Source.3            6438
Detailed Purpose    6308
Source.2            5732
Comments            5478
dtype: int64

Top missing columns in kelvins_labels_clean:
originator_id         0
area_to_mass_ratio    0
feature_2             0
feature_3             0
semi_major_axis_km    0
eccentricity          0
inclination_deg       0
mean_anomaly_deg      0
raan_or_arg_deg       0
dtype: int64
```

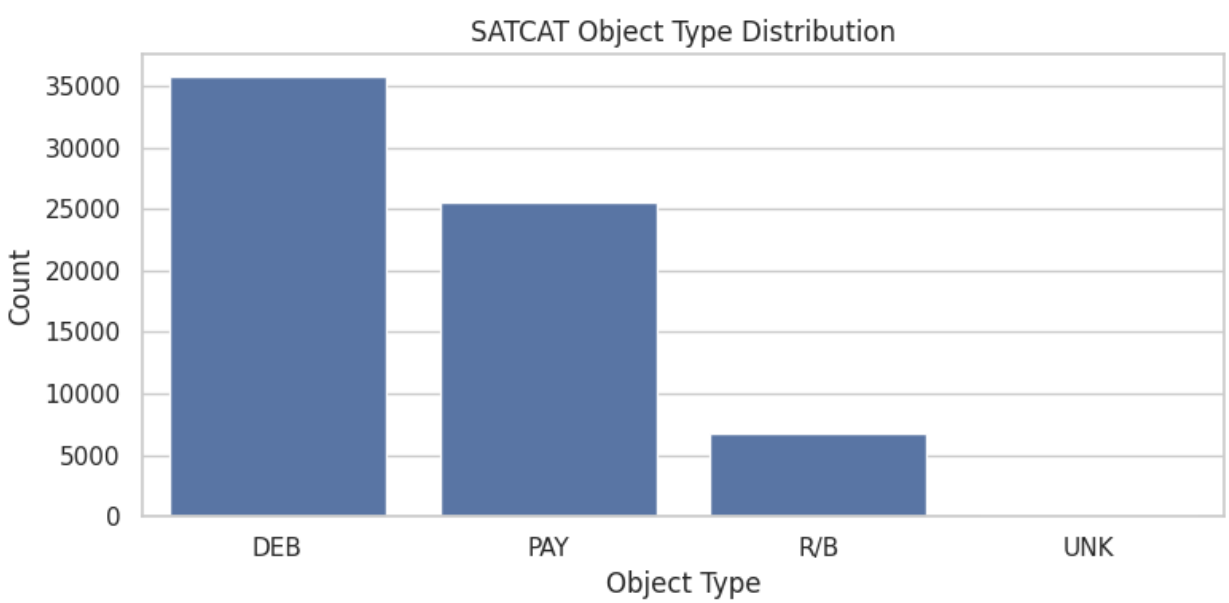
SATCAT Analysis

Summarize object composition and temporal launch trend.

```
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

# Object type distribution
obj_counts =
satcat_clean["OBJECT_TYPE"].fillna("UNKNOWN").value_counts().reset_ind
ex()
obj_counts.columns = ["OBJECT_TYPE", "count"]

display(obj_counts)

plt.figure(figsize=(8, 4))
sns.barplot(data=obj_counts, x="OBJECT_TYPE", y="count")
plt.title("SATCAT Object Type Distribution")
plt.xlabel("Object Type")
```

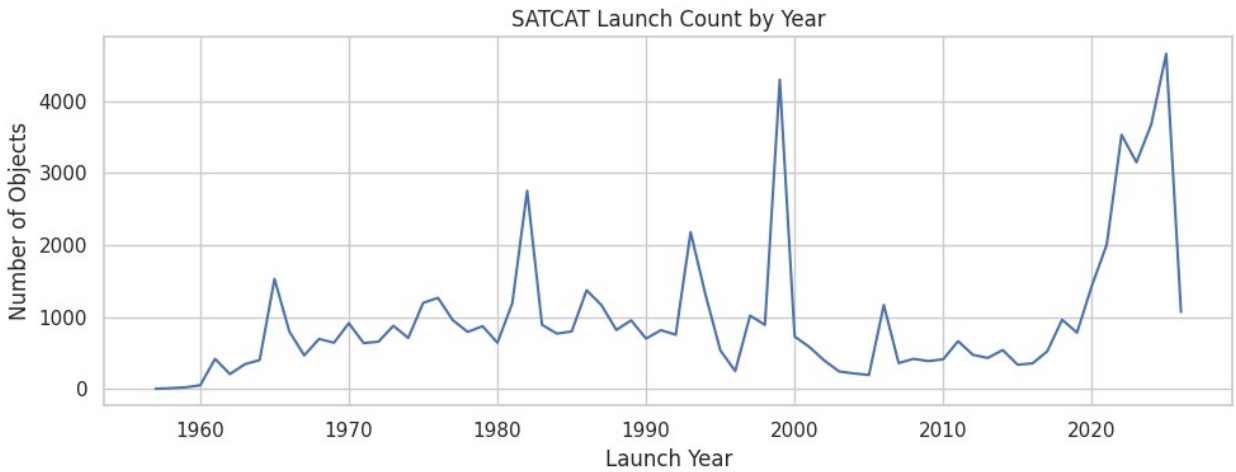
```
plt.ylabel("Count")
plt.tight_layout()
plt.show()

# Launch trend by year
launch_year_counts = (
    satcat_clean.dropna(subset=["LAUNCH_DATE"])
    .assign(launch_year=satcat_clean["LAUNCH_DATE"].dt.year)
    .groupby("launch_year")
    .size()
    .reset_index(name="count")
)

display(launch_year_counts.tail(15))

plt.figure(figsize=(10, 4))
sns.lineplot(data=launch_year_counts, x="launch_year", y="count")
plt.title("SATCAT Launch Count by Year")
plt.xlabel("Launch Year")
plt.ylabel("Number of Objects")
plt.tight_layout()
plt.show()
```

```
  OBJECT_TYPE  count
0         DEB  35750
1         PAY  25588
2         R/B   6828
3         UNK    164
```




```
    launch_year  count
55         2012    474
```

```
56          2013    432
57          2014    542
58          2015    338
59          2016    355
60          2017    524
61          2018    963
62          2019    783
63          2020   1436
64          2021   2006
65          2022   3534
66          2023   3151
67          2024   3680
68          2025   4659
69          2026   1070
```



UCS Analysis

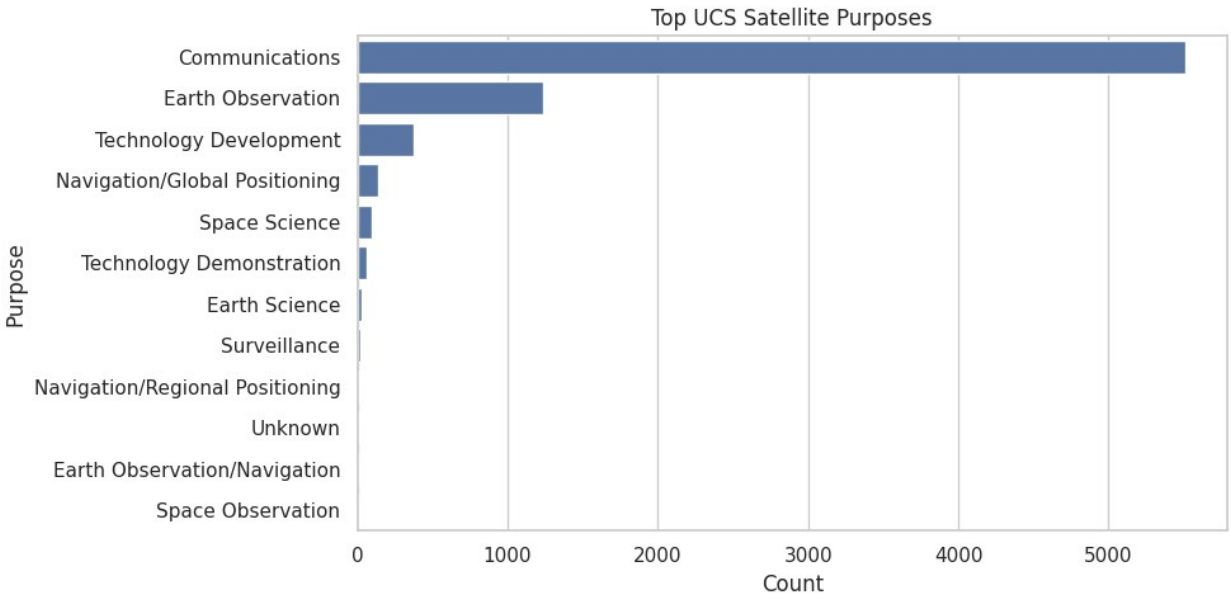
Summarize operational satellite purpose, orbit class, and launch trend.

```
ucs_purpose = (

ucs_clean["Purpose"].fillna("Unknown").value_counts().reset_index()
)
ucs_purpose.columns = ["Purpose", "count"]
display(ucs_purpose.head(15))

plt.figure(figsize=(10, 5))
sns.barplot(data=ucs_purpose.head(12), x="count", y="Purpose")
plt.title("Top UCS Satellite Purposes")
plt.xlabel("Count")
plt.ylabel("Purpose")
plt.tight_layout()
plt.show()
```

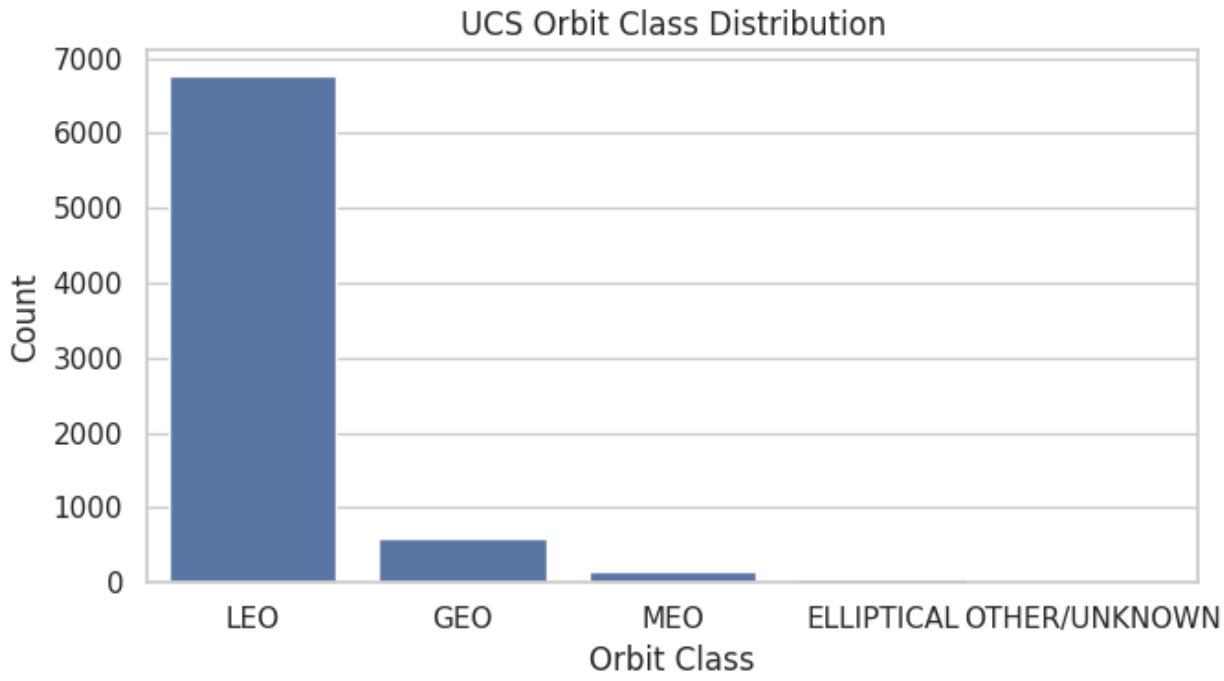
```
orbit_norm = (
    ucs_clean["Class of Orbit"]
    .astype(str)
    .str.strip()
    .str.upper()
    .replace({"LEO": "LEO", "LEO ": "LEO", "LEOO": "LEO", "LEO":
"LEO", "LEO": "LEO"})
)
orbit_norm = orbit_norm.where(orbit_norm.isin(["LEO", "MEO", "GEO",
"ELLIPTICAL"]), "OTHER/UNKNOWN")
ucs_orbit = orbit_norm.value_counts().reset_index()
ucs_orbit.columns = ["Class of Orbit", "count"]
display(ucs_orbit)

plt.figure(figsize=(7, 4))
sns.barplot(data=ucs_orbit, x="Class of Orbit", y="count")
plt.title("UCS Orbit Class Distribution")
plt.xlabel("Orbit Class")
plt.ylabel("Count")
plt.tight_layout()
plt.show()

ucs_launch_year = (
    ucs_clean.dropna(subset=["Date of Launch"])
    .assign(launch_year=lambda d: d["Date of Launch"].dt.year)
)
ucs_launch_year = ucs_launch_year[
    (ucs_launch_year["launch_year"] >= 1957) &
(ucs_launch_year["launch_year"] <= 2035)
]
ucs_launch_year = (

ucs_launch_year.groupby("launch_year").size().reset_index(name="count"
)
)

display(ucs_launch_year.tail(15))

plt.figure(figsize=(10, 4))
sns.lineplot(data=ucs_launch_year, x="launch_year", y="count")
plt.title("UCS Launch Count by Year")
plt.xlabel("Launch Year")
plt.ylabel("Number of Satellites")
plt.tight_layout()
plt.show()
```

```
                                     Purpose  count
0                            Communications   5514
1                         Earth Observation   1235
2                     Technology Development    372
```

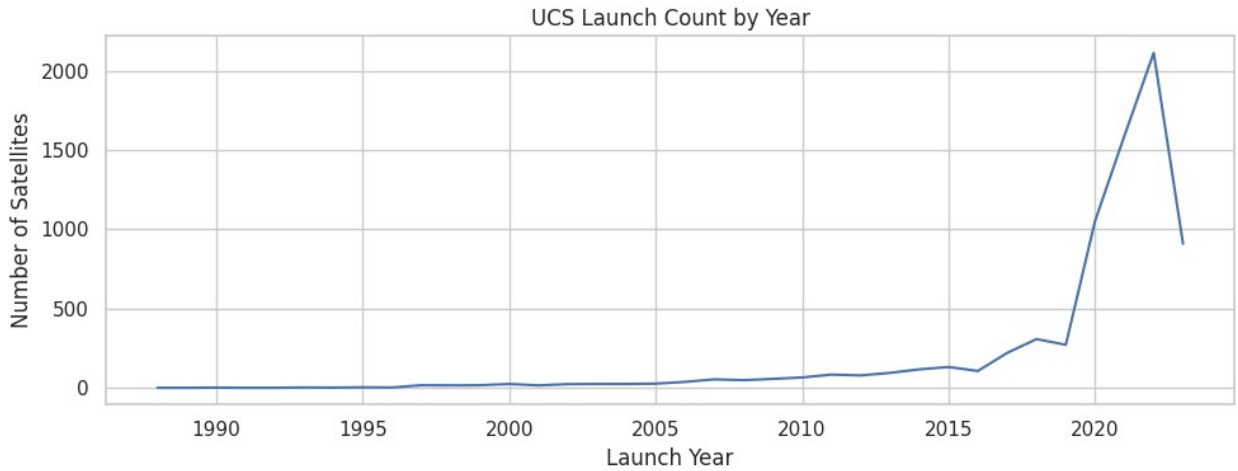
```
3              Navigation/Global Positioning    142
4                              Space Science     99
5                   Technology Demonstration     64
6                              Earth Science     28
7                               Surveillance     20
8            Navigation/Regional Positioning     13
9                                    Unknown     12
10              Earth Observation/Navigation      9
11                          Space Observation      9
12  Earth Observation/Technology Development      7
13                            Meteorological      6
14          Communications/Maritime Tracking      5
```




```
  Class of Orbit  count
0            LEO   6768
1            GEO    589
2            MEO    143
3     ELLIPTICAL     59
4  OTHER/UNKNOWN      4
```




```
    launch_year  count
21         2009     57
22         2010     66
23         2011     84
24         2012     79
25         2013     95
26         2014    117
27         2015    132
28         2016    107
29         2017    221
30         2018    308
31         2019    272
32         2020   1054
33         2021   1590
34         2022   2113
35         2023    909
```



Kelvins Analysis

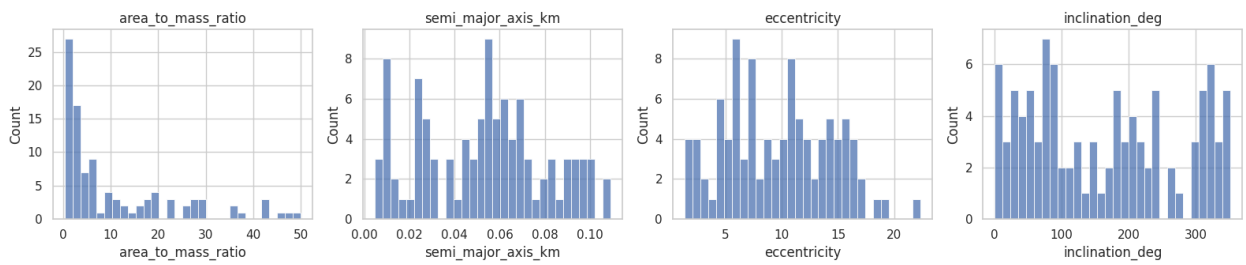
Debris-vs-satellite class balance and basic orbital element distributions.

```
kelvins_numeric = kelvins_labels_clean.apply(pd.to_numeric,
errors="coerce")
kelvins_stats = kelvins_numeric.describe().T
kelvins_stats["missing_ratio"] = kelvins_numeric.isna().mean()
display(kelvins_stats)

kelvins_plot_cols = [
    "area_to_mass_ratio",
    "semi_major_axis_km",
    "eccentricity",
    "inclination_deg",
]
kelvins_plot_cols = [c for c in kelvins_plot_cols if c in
kelvins_numeric.columns]

if kelvins_plot_cols:
    fig, axes = plt.subplots(1, len(kelvins_plot_cols), figsize=(4 *
len(kelvins_plot_cols), 3.5))
    if len(kelvins_plot_cols) == 1:
        axes = [axes]
    for ax, col in zip(axes, kelvins_plot_cols):
        sns.histplot(kelvins_numeric[col].dropna(), bins=30,
kde=False, ax=ax)
        ax.set_title(col)
    plt.tight_layout()
    plt.show()

trajectory_summary = pd.DataFrame(
    [
        {
```

```
            "subset": "deb_train",
            "rows": len(kelvins_deb_train),
            "trajectories":
kelvins_deb_train["trajectory_id"].nunique(),
            "avg_points_per_trajectory": round(len(kelvins_deb_train)
/ max(kelvins_deb_train["trajectory_id"].nunique(), 1), 2),
        },
        {
            "subset": "deb_test",
            "rows": len(kelvins_deb_test),
            "trajectories":
kelvins_deb_test["trajectory_id"].nunique(),
            "avg_points_per_trajectory": round(len(kelvins_deb_test) /
max(kelvins_deb_test["trajectory_id"].nunique(), 1), 2),
        },
        {
            "subset": "sat",
            "rows": len(kelvins_sat),
            "trajectories": kelvins_sat["trajectory_id"].nunique(),
            "avg_points_per_trajectory": round(len(kelvins_sat) /
max(kelvins_sat["trajectory_id"].nunique(), 1), 2),
        },
    ]
)
display(trajectory_summary)
```

```
                     count          mean          std          min  \
originator_id        100.0     49.700000    29.211005      1.00000
area_to_mass_ratio   100.0     10.983837    12.867690      0.31969
feature_2            100.0  -3604.300000  2147.024370  -7295.00000
feature_3            100.0  42254.820000     3.662704  42247.85000
semi_major_axis_km   100.0      0.052836     0.027862      0.00459
eccentricity         100.0      9.644671     4.730841      1.35230
inclination_deg      100.0    163.251015   111.376679      0.63820
mean_anomaly_deg     100.0    189.424480    95.798442      2.22560
raan_or_arg_deg      100.0     90.134697    91.359688      4.41410

                              25%           50%           75%
max  \
originator_id          21.750000     49.500000     75.250000
100.00000
area_to_mass_ratio      1.764633      4.989640     18.281610
49.92149
feature_2           -5220.000000  -3565.000000  -1872.500000     -
5.00000
feature_3           42251.757500  42254.700000  42257.900000
42262.44000
semi_major_axis_km      0.027844      0.055848      0.070015
0.10909
eccentricity            5.992275      9.588400     13.396050
```

```
22.33870
inclination_deg        68.921200    152.959700    247.065700
351.24760
mean_anomaly_deg      105.347325    190.396350    263.455075
356.51910
raan_or_arg_deg        32.582625     59.556000    102.013650
355.28980

                    missing_ratio
originator_id                 0.0
area_to_mass_ratio            0.0
feature_2                     0.0
feature_3                     0.0
semi_major_axis_km            0.0
eccentricity                  0.0
inclination_deg               0.0
mean_anomaly_deg              0.0
raan_or_arg_deg               0.0
```




```
      subset    rows  trajectories  avg_points_per_trajectory
0  deb_train     751           100                       7.51
1   deb_test     751           100                       7.51
2        sat  109600           100                    1096.00
```

Integrated Findings

Final EDA summary and checks for modeling-ready artifacts.

```
summary_points = pd.DataFrame(
    [
        {
            "dataset": "SATCAT",
            "rows": len(satcat_clean),
            "notes": "Large historical object catalog; object types
and launch trends available",
        },
        {
            "dataset": "UCS",
            "rows": len(ucs_clean),
            "notes": "Operational context with purpose and orbit class
```

```
metadata",
        },
        {
            "dataset": "Kelvins",
            "rows": len(kelvins_labels_clean),
            "notes": "Labeled orbital features plus long-form
trajectory files",
        },
    ]
)

display(summary_points)

expected_outputs = [
    PROCESSED_DIR / "satcat_clean.csv",
    PROCESSED_DIR / "ucs_clean.csv",
    PROCESSED_DIR / "kelvins_labels_clean.csv",
    PROCESSED_DIR / "kelvins_deb_train_long.csv",
    PROCESSED_DIR / "kelvins_deb_test_long.csv",
    PROCESSED_DIR / "kelvins_sat_long.csv",
]

outputs_status = pd.DataFrame(
    {
        "file": [p.name for p in expected_outputs],
        "exists": [p.exists() for p in expected_outputs],
    }
)
display(outputs_status)

print("EDA complete: cleaned artifacts are ready under
input_data/processed/")
```

```
   dataset   rows                                                notes
0   SATCAT  68330  Large historical object catalog; object types ...
1      UCS   7563  Operational context with purpose and orbit cla...
2  Kelvins    100  Labeled orbital features plus long-form trajec...
```

```
                          file  exists
0            satcat_clean.csv    True
1               ucs_clean.csv    True
2    kelvins_labels_clean.csv    True
3  kelvins_deb_train_long.csv    True
4   kelvins_deb_test_long.csv    True
5        kelvins_sat_long.csv    True
```

```
EDA complete: cleaned artifacts are ready under input_data/processed/
```